Arizona State University

DECODING THE GRID: AN EXPLORATORY ANALYSIS OF TRAFFIC AND ACCIDENT PATTERNS IN THE CITY OF CHICAGO

Fall 2025 Capstone Project

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Introduction

A brief look into the motor vehicle accidents in the United States

- Motor vehicle crashes are a critical public health and economic challenge
  - 40,901 people died in 2023
  - Roadway fatalities increased post-pandemic, with 2021 recording the highest number since 2005 (43,230 deaths)
  - Economic impact on Americans totals $340 billion annually
  - Speeding was a factor in 29% of motor vehicle crash deaths in 2023
- Many municipalities use Automated Traffic Enforcement (ATE), such as speed and red-light cameras
  - NHTSA assigns this the highest rating for effectiveness
  - Mixed findings create a need for understanding local impacts
- Recent research has had success in predicting crash severity by utilizing various machine learning methods

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Project Goal and Objectives

Goal:

- To present an encompassing, comprehensive analysis of traffic accident data from the City of Chicago (CoC) to advance understanding of urban traffic safety

Objectives:

- Evaluate the effectiveness of Chicago's automated enforcement infrastructure (speed and red-light cameras) on accident frequency and injury severity at the individual camera level
- Investigate the relationship between neighborhood poverty rates and traffic safety outcomes by joining crash records with Census data
- Develop an interactive geospatial visualization to dynamically identify accident hotspots using unsupervised clustering algorithms
- Develop machine learning models to predict accident injury severity based on environmental, temporal, spatial, and roadway characteristics

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Data Sources and Preprocessing

Primary Data Source	City of Chicago (CoC) open data portal. Traffic Crash Records (911K rows, Sep 2017 - Oct 2025). Red Light and Speeding Camera Locations (360 rows, 2004 - 2025) and Violations (1.5M rows, 2014 – 2025)
Secondary Data Source	U.S Census Bureau 5-Year American Community Survey (ACS). Contains socioeconomic variables for 61 zip codes (2023 data)

- Python/Pandas for column renaming, typecasting, categorical conversion, and null imputation/removal. Spatial analysis was applied to implement zip codes for each coordinate data point
- Incorporated shapefiles for zip code boundaries. Obscure census variable codes were renamed into human-readable columns to ensure accurate mapping of socioeconomic factors to crash locations
- Manual mapping was required to link respective intersections in the violation logs to the camera locations dataset.

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Redlight and Speeding Camera Analysis

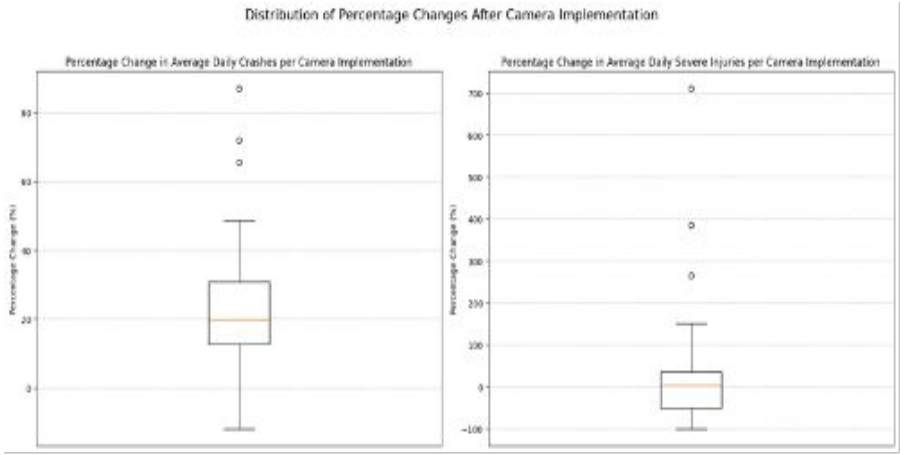
A deep dive into the impact of Camera Placement on Crash and Severe Injury Occurrence

Methodology:

- A quarter-mile buffer radius was established around each camera location. Incident data were segmented into pre- and post-implementation cohorts based on the camera's "go-live" date. Rates were standardized by calculating daily incident rates
- The Wilcoxon Signed-Rank Test was used to determine the statistical significance in the difference in crash and severe injury rates pre- and post-camera implementation

Results:

- The Wilcoxon Signed-Rank Test yielded a p-value of 0.8402. We failed to reject the null hypothesis, indicating there was no statistically significant difference in severe injury rates pre- and post-implementation for the analyzed cameras
- The analysis resulted in a p-value of approximately 0.00. The null hypothesis was rejected, confirming a statistically significant increase in daily average crashes post-implementation. Almost every camera pair analyzed exhibited an increase in daily average crashes



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Poverty Rate Association Analysis

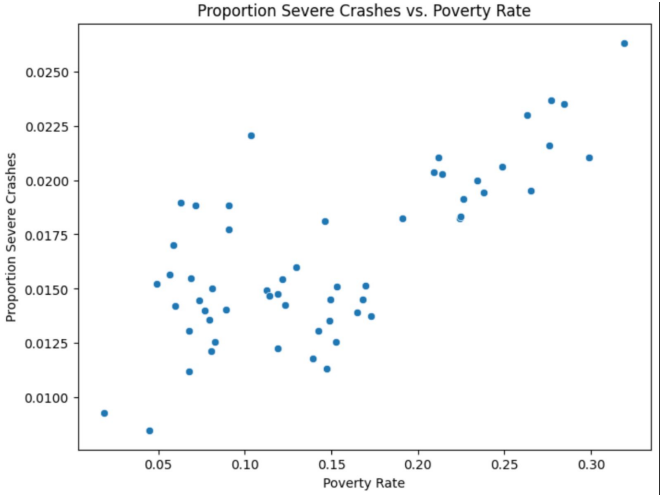
How does poverty relate to crash severity

Data Preparation: Merged Crash + Census Data at ZIP Code Level. Normalized inputs (proportions/averages) to control for volume. Variables derived:

- **Outcome:** proportion_severe_crashes (Fatal/Incapacitating)
- **Key Predictor:** poverty_rate (% of residents below the poverty line).
- **Infrastructure Controls:** proportion_poor_road_cond and proportion_inadequate_lighting.
- **Traffic Controls:** proportion_vru_crashes (pedestrians/cyclists) and average_speed_limit.

Statistical Analysis: The analysis was conducted using ordinary least squares (OLS) regression. Bivariate and nested multivariate regression models were used to examine the correlations between predictor variables and outcome variable. Three models were built to test stability.

- **Model 1:** Tested poverty rate alone.
- **Model 2:** Added Infrastructure (Poor Road Condition, Inadequate Lighting) to see if they changed the effect of poverty.
- **Model 3:** Added Traffic Features (VRU, Speed Limit) to control for everything at once.



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Poverty Rate Association Analysis

How does poverty relate to accident severity

Key Findings:

- The poverty rate exhibited a very strong positive association ($r = +0.724$, $p < 0.001$) with the proportion of severe crashes. The effect remained highly significant across all model specifications. Implying that in real world, a ~1.4% point difference in severe crash rates exists between the wealthiest and poorest neighborhoods—a substantial disparity given the city-wide average is only ~1.5–2%
- Initial bivariate tests showed that poor road conditions ($r = +0.361$) and inadequate lighting ($r = +0.396$) were linked to a higher proportion of severe crashes. However, these effects were eliminated ($p > 0.05$) once poverty rate was controlled for, indicating that poverty fully mediates these infrastructure issues.
- Areas involved with a higher proportion of pedestrian and bicyclist crashes showed significantly lower crash severity ($r = –0.406$, $p = 0.032$). It suggests mixed traffic environments may foster lower operating speeds or different crash typologies that reduce fatal outcomes.
- The average speed limit exhibited no meaningful independent association with the proportion of severe crashes at the ZIP-code level ($p = 0.075$).

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Spatio Temporal Hotspot Analysis

A when(spatio) and where(temporal) analysis of crash data

Goal : Identify where and when crashes happen to improve resource allocation.

- **The Approach:** Used **K-Means Clustering** to group spatially dense crash points.

  Why K-Means? Unlike density-based methods (e.g., DBSCAN), K-Means enforces a fixed number of clusters, ensuring the stability needed for smooth, frame-by-frame animation.

- **The Input:** Chicago crash data (2018–Present)
- **The Visualization Logic:**
  - **Marker Size:** Crash Frequency (*How many?*)
  - **Marker Color:** Injury Severity (*How bad?*)
- **The Implementation :** Built in **Python** using **Pandas** for data processing and **Plotly** to generate the interactive, frame-based animation.

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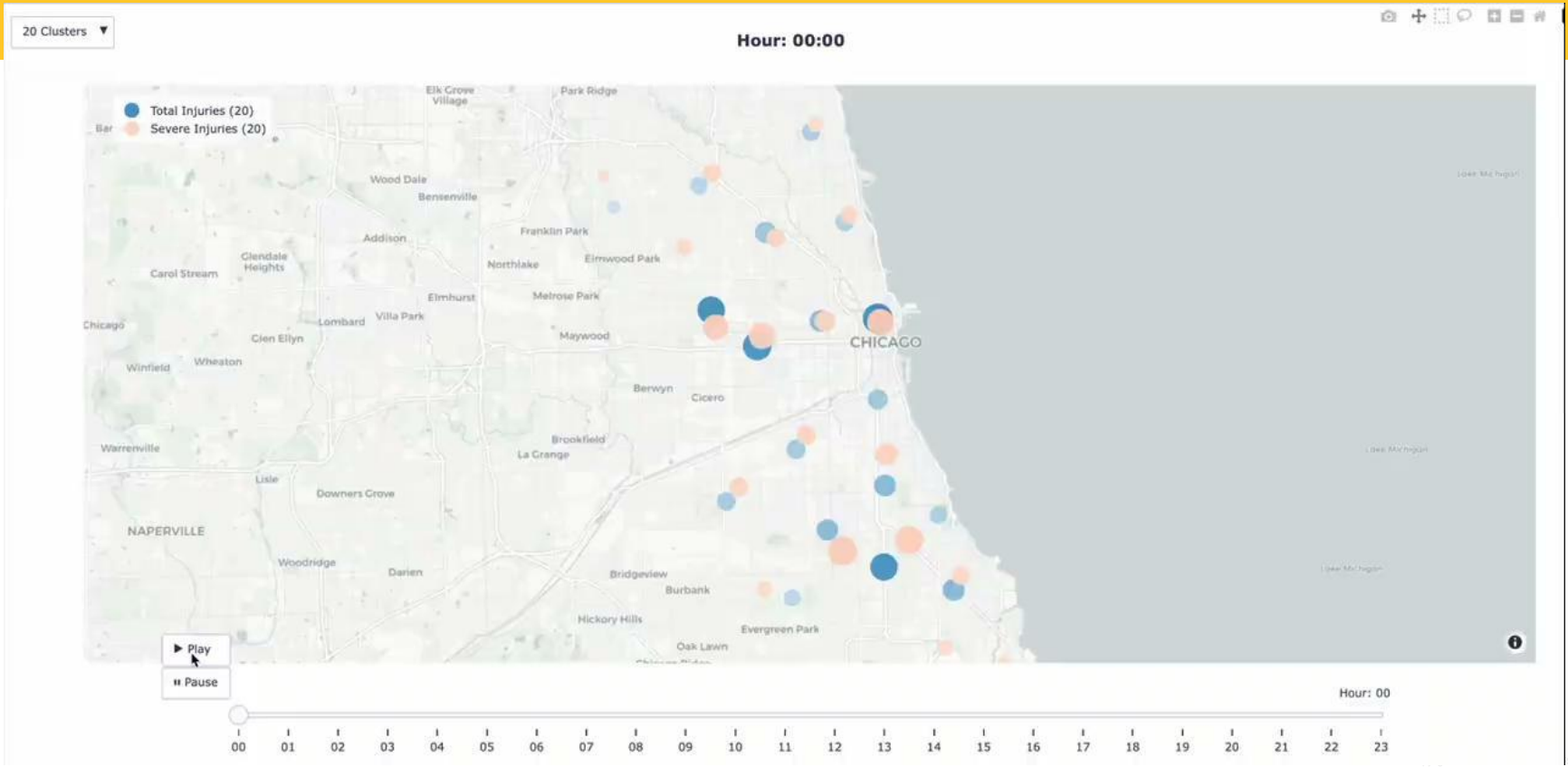
Spatio Temporal Hotspot Analysis

Map showing Total Injuries vs Severe Injuries



Spatio Temporal Hotspot Analysis

Weekend vs Weekday crashes

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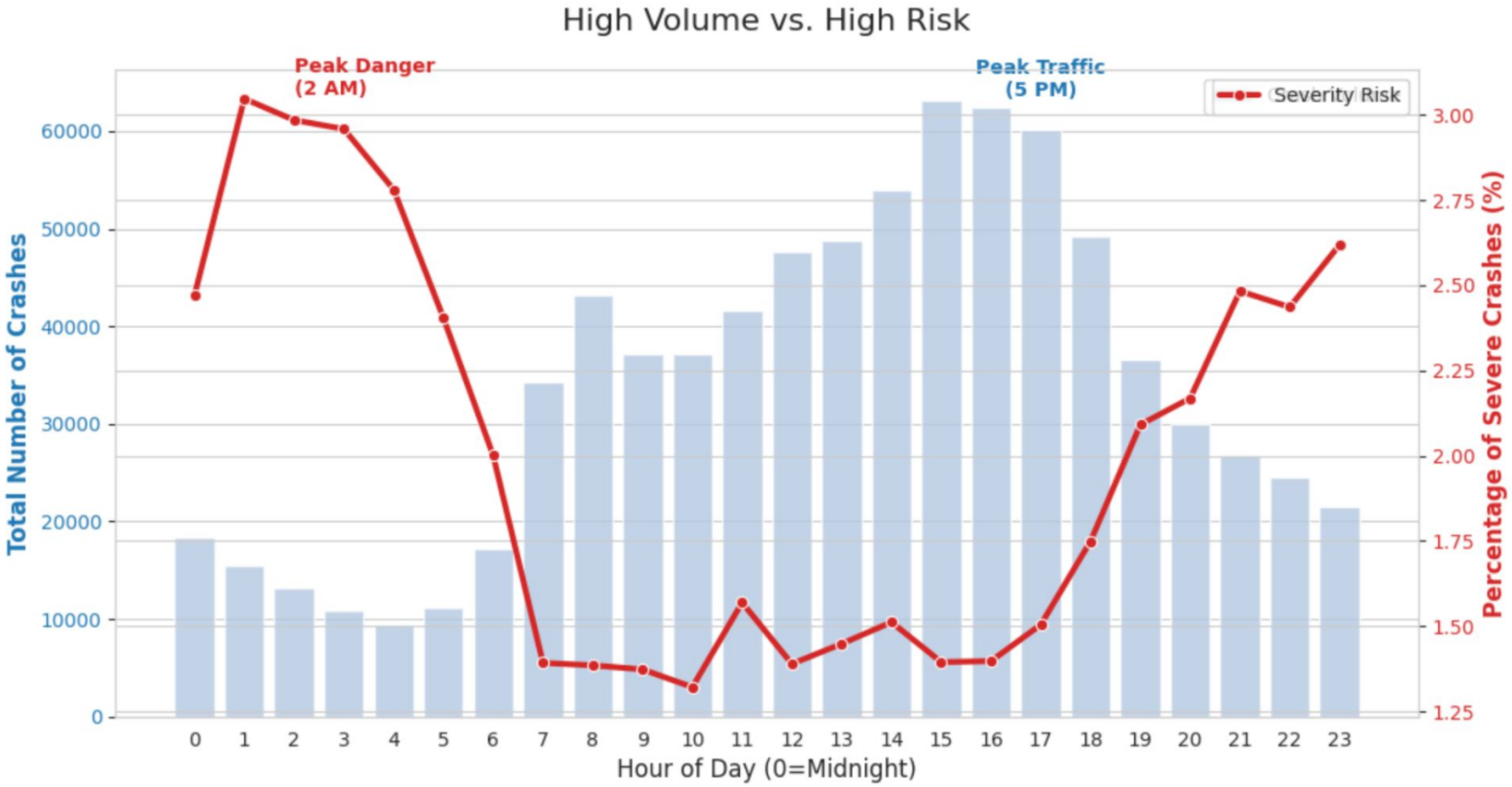
Spatio Temporal Hotspot Analysis



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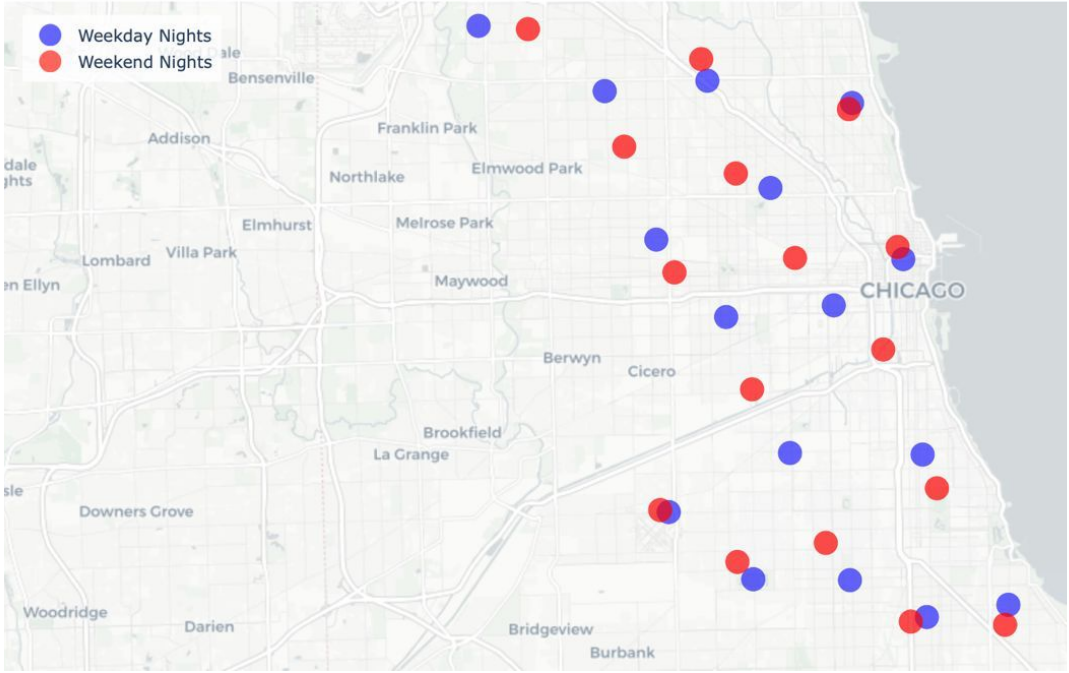
Spatio Temporal Hotspot Analysis



Geographic Migration (Commute vs. Culture): The map reveals a distinct behavioral shift: Weekday hotspots (**Blue**) align with major highways (I-90/94) driven by logistics and commuting, while Weekend hotspots (**Red**) migrate off the expressways and into dense **entertainment districts** like River North and Wicker Park.

Dynamic Resource Allocation: This spatial separation proves that safety resources cannot remain static. To reduce severe outcomes, Police and EMS units must **physically redeploy** from commuter routes to neighborhood intersections during weekend "nightlife" hours.

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Predictive Modeling

Predicting Severe Traffic Crashes: Problem & Data Challenges

- Goal: Predict whether a reported crash is **severe (injury/fatal)** soon after it occurs.
- Dataset size: **~950k records**, severe crash rate only **1.66%**.
- Key challenge: **Extreme class imbalance** + overlapping patterns.
- Accuracy is misleading → **PR-AUC, Recall, F2-score** used instead of ROC/Accuracy.
- Why PR-AUC? More sensitive to rare severe cases (ROC overestimates performance).

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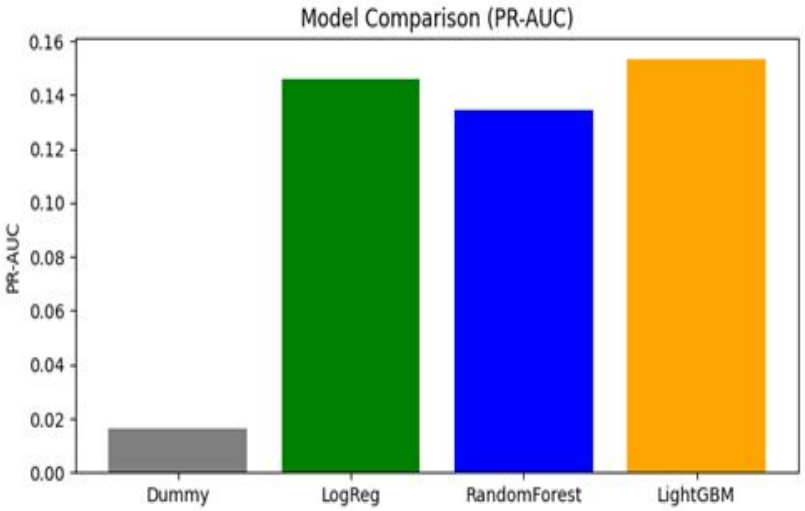
Baseline vs Advanced Models

Model Development: From Baselines to LightGBM

Baselines:

- Dummy → Accuracy ~98.4% but **0 severe cases predicted** → PR-AUC ≈ 0.016.
- Logistic Regression → PR-AUC **0.146** (linear patterns detected).
- Random Forest → PR-AUC **0.134** (non-linear but not better).



Advanced Model:

- LightGBM selected (best for large tabular imbalanced data).
- Achieved **PR-AUC ~0.153**.
- Hyperparameter tuning, class weights & resampling, **minimal improvement**.

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Class Imbalance Solutions & Threshold Tuning

Handling Imbalance & Improving Recall

Tested:

- Class weights (balanced vs none)

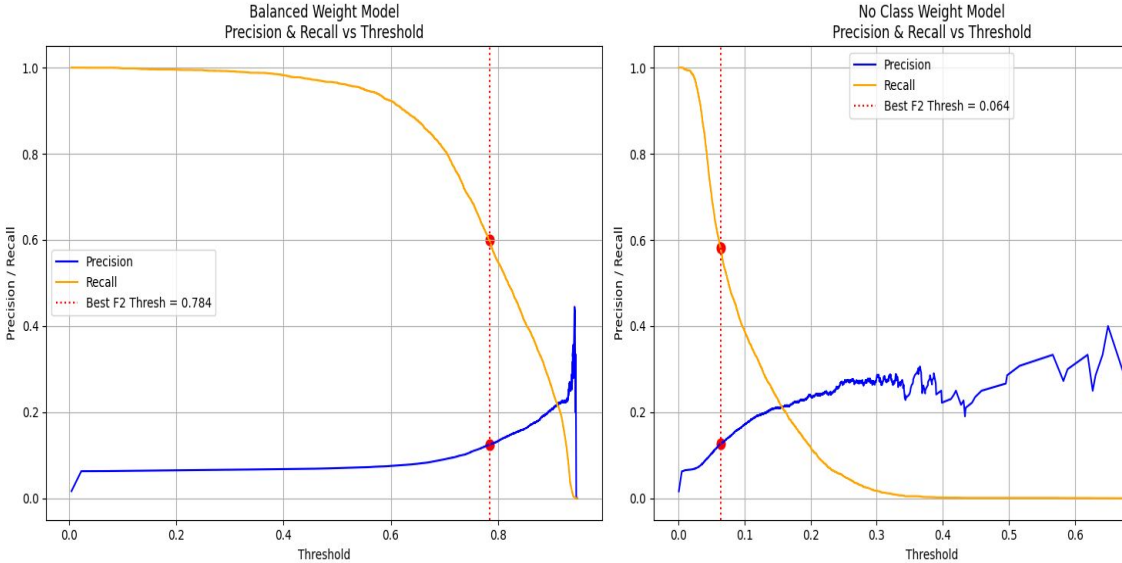
Result: Models remained **very similar** → LightGBM already learns good probabilities.

Breakthrough: Threshold tuning using **F_2-score** (recall-focused).

- Default threshold 0.50 → recall ≈ 0%.
- Optimal threshold **0.061** →
- recall **55%+**, precision **12–13%**.

Interpretation:

The system captures more than half of severe crashes



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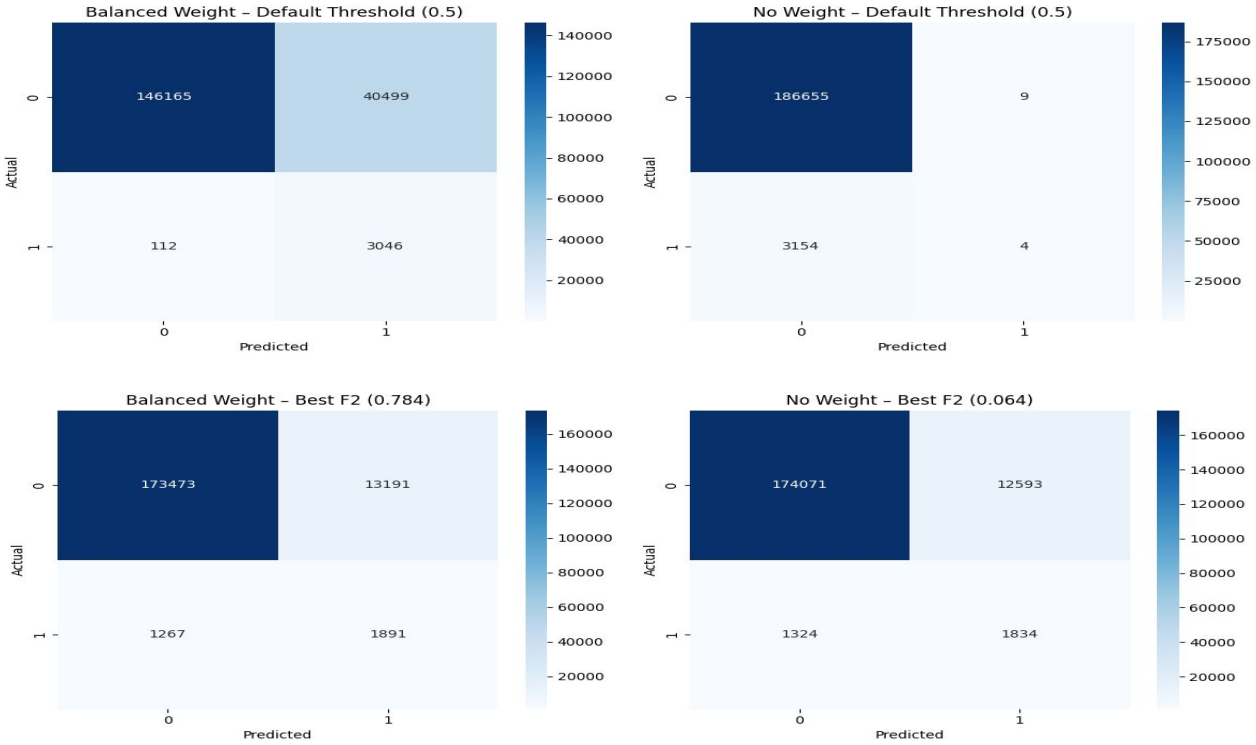
Default vs F2 Optimized Threshold(recall focus)

Confusion Matrix Result



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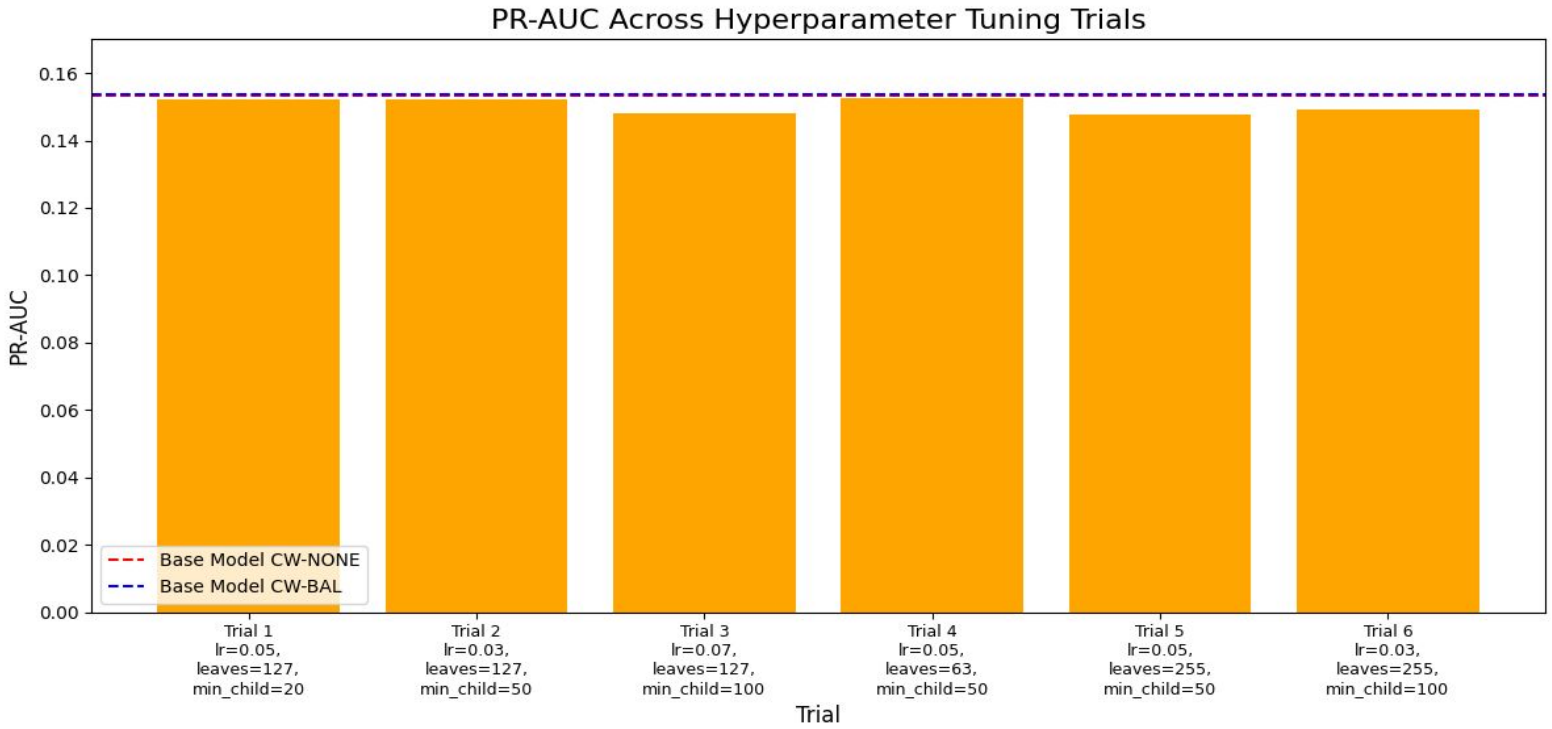
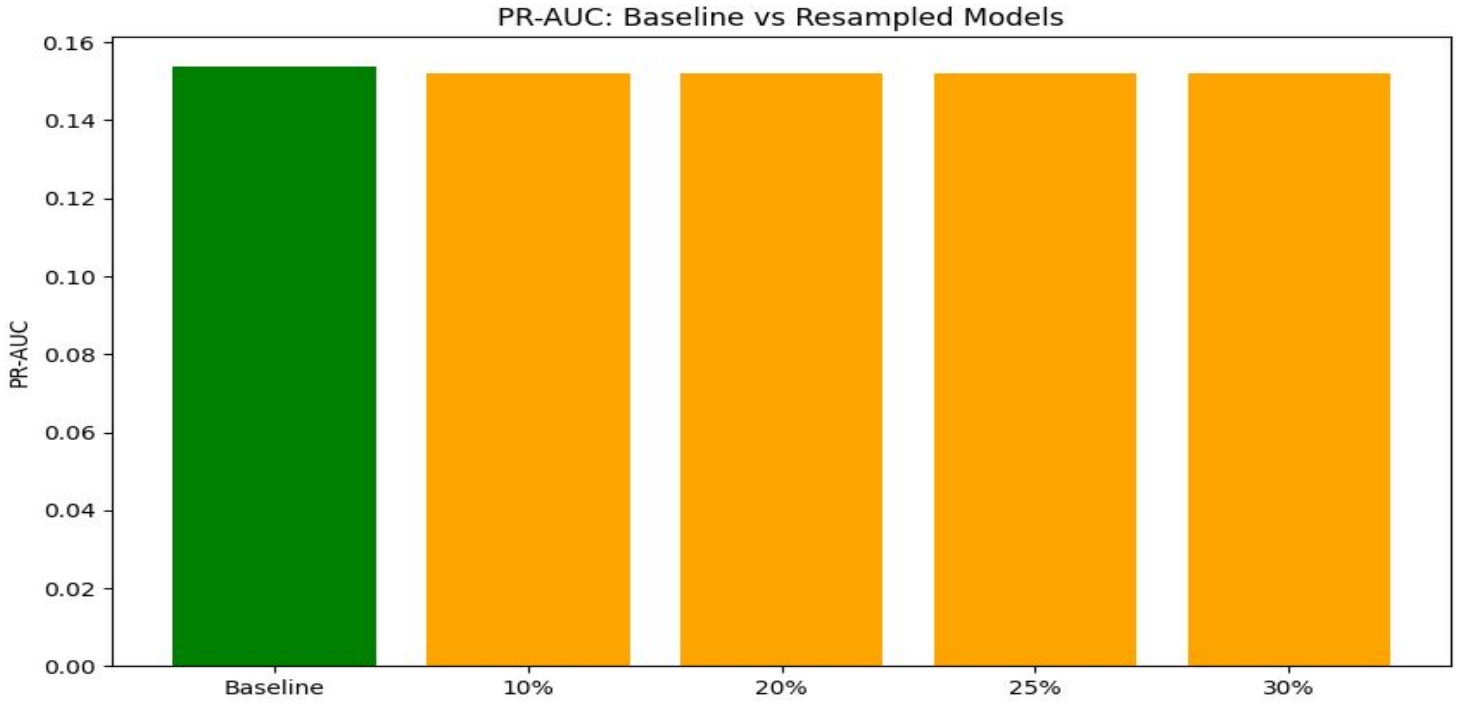
Baseline vs HyperParameter Trails

Model Comparison Results



Baseline vs Resampling

Model Comparison Results



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Insights, Interpretability & Conclusion

Model Insights & Final Outcome

Feature Importance:

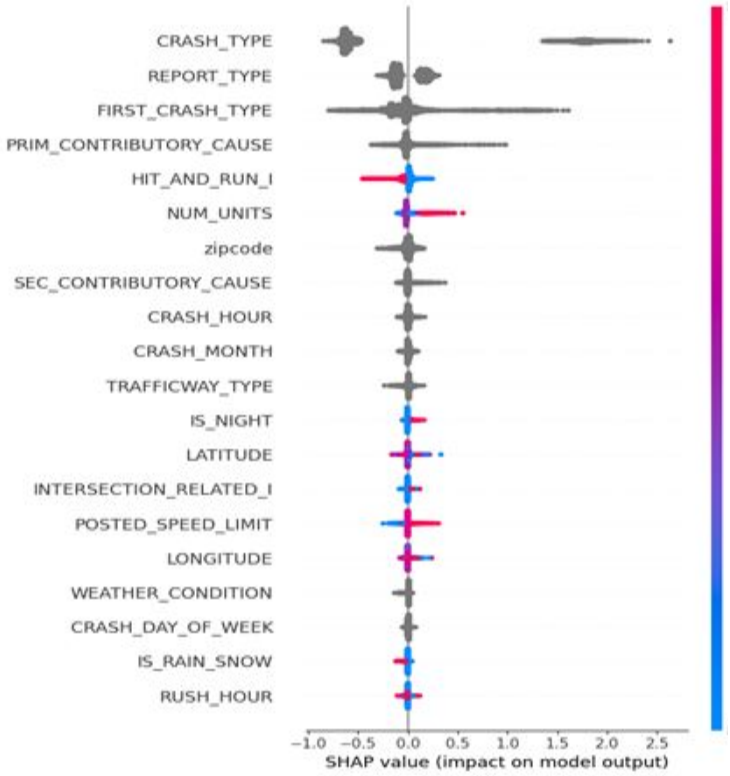
- Crash type, contributory cause, units involved, hit-and-run,hour of crash.
- Spatial & environmental factors also relevant.

SHAP Analysis:

- Confirms model uses logical features aligned with road-safety patterns.

Final Conclusion:

- Predicting severe crashes is **feasible but challenging**.
- LightGBM performs best but **threshold tuning** is the real improvement.



Conclusion and Recommendations

- Red light and speeding camera analysis showed no significant indicators
  - Data showed frequency actually increased, likely due to distribution of data
  - Fixed effects regression controlling for data may yield better results
- Hotspot analysis provides useful insight into time-specific trends
  - This method is developed can be directly used by transportation/law enforcement
  - Next steps could account for travel distance as opposed to basic line distanc
- Predictive ML modeling showed some promise and identified significant elements contributing to accident severity
  - Location and accident circumstance matter most
  - More extensive modeling could be done to incorporate vehicle and driver factors
- Poverty association analysis showed a strong correlation
  - Intermediary elements are still largely unproven
  - Further research on the root of the causal relationship is necessary

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Questions?

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THANK YOU!

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